TASK 1 REPORT

DNA/Protein Sequence Analysis of BRCA1 Homologs in Three Primate Species

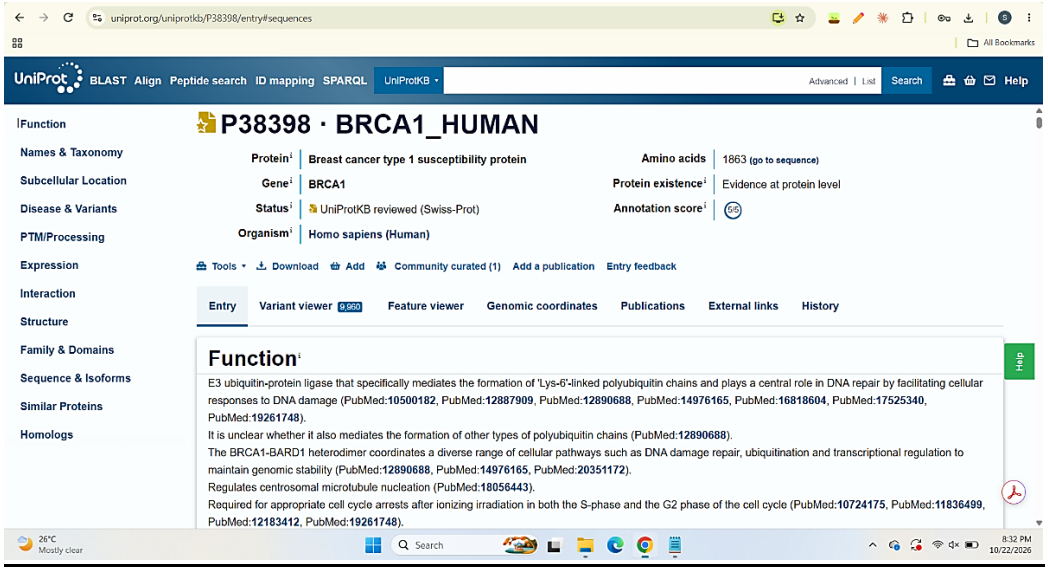
INTRODUCTION

Breast cancer type 1 susceptibility protein (BRCA1) is an important protein involved in the maintenance of genomic stability and DNA damage response. The BRCA1 protein participates in processes such as DNA repair and regulation of cellular responses to DNA damage. Because BRCA1 is functionally important, its sequence is highly conserved among closely related species.

In this study, protein sequences of BRCA1 homologs from three primate species were analyzed using the NCBI BLAST (Basic Local Alignment Search Tool) program. The three species selected were chimpanzee (*Pan troglodytes*), gorilla (*Gorilla gorilla gorilla*), and Sumatran orangutan (*Pongo abelii*).

OBJECTIVES

- To obtain BRCA1 protein sequences from NCBI.
- To perform BLAST protein sequence analysis.
- To identify homologous BRCA1 sequences in three primate species.
- To compare the percentage identity and similarity between the sequences.
- To analyze the BLAST alignment score, E-value, and gaps.
- To determine the degree of conservation of BRCA1 among the selected primates.



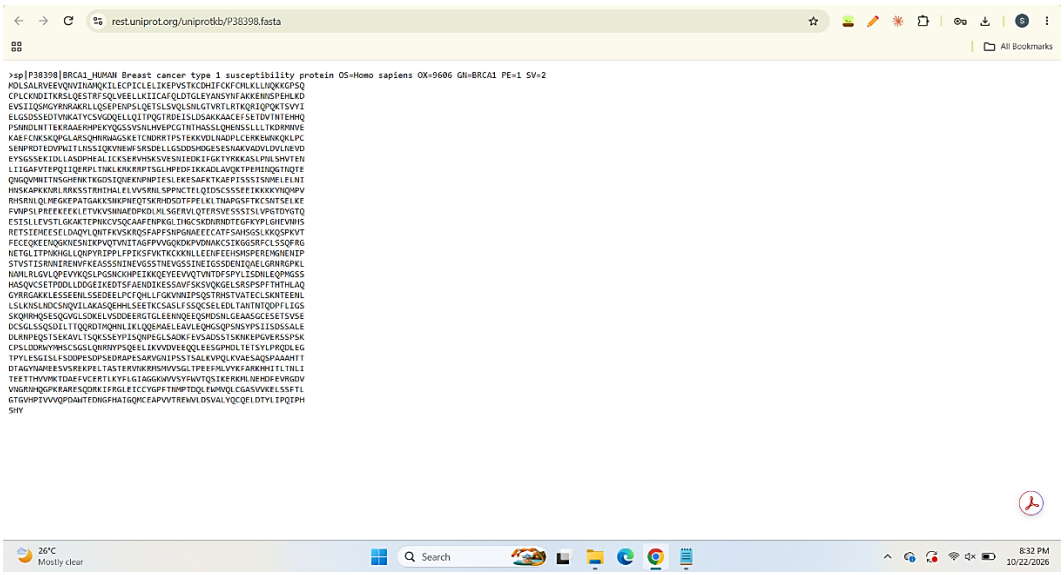
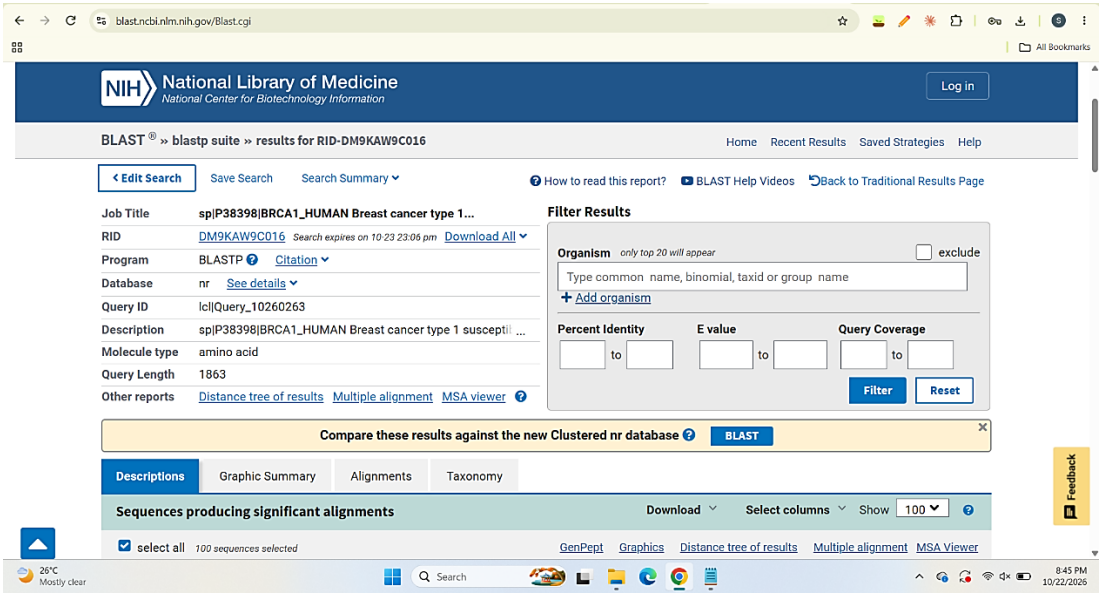


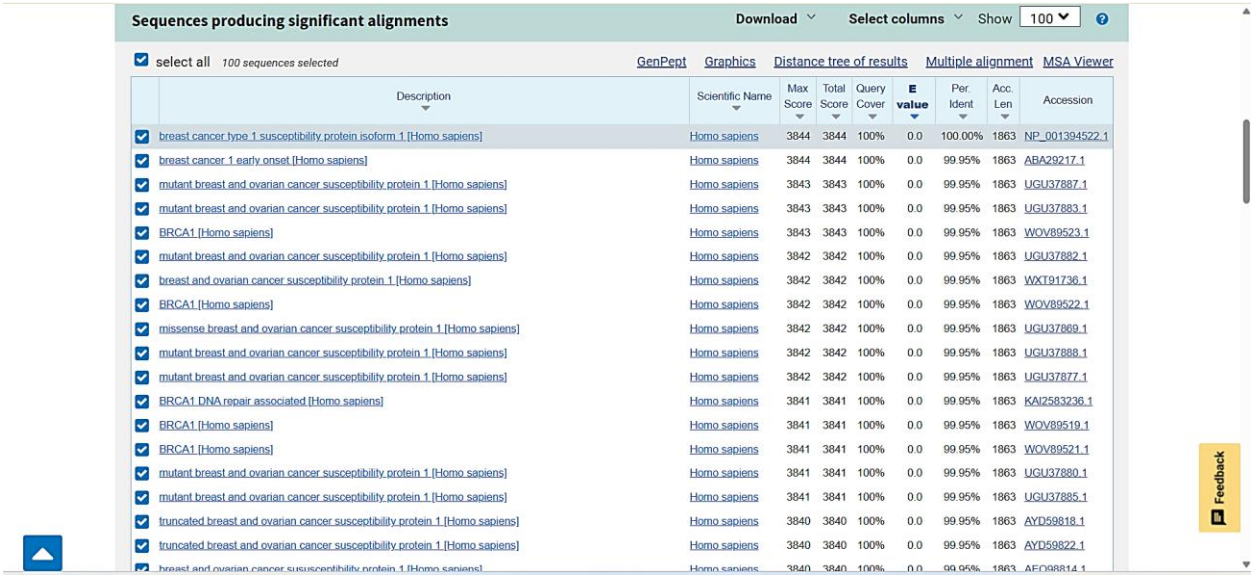
Fig : FASTA sequence of BRCA-1

METHODOLOGY

The BRCA1 protein sequence data were obtained from the National Center for Biotechnology Information (NCBI) database. Protein sequences from chimpanzee, gorilla, and Sumatran orangutan were selected for comparative analysis.

The NCBI BLASTP program was used to compare the protein sequences and identify homologous sequences.





The BLAST results were evaluated using the following parameters:

- **Sequence identity:** Percentage of amino acid residues that are exactly the same between the two sequences.
- **Positives:** Percentage of identical or conservatively substituted amino acids.
- **Bit score:** A measure of the quality of the alignment; a higher score indicates a stronger alignment.
- **E-value:** Indicates the probability of obtaining an alignment of similar quality by chance. An E-value of 0.0 represents an extremely significant match.
- **Gaps:** Positions where residues have been inserted or deleted in the alignment.

The complete alignment covered positions 1–1863, meaning that the entire length of each protein was aligned.

SEQUENCES SELECTED

The BRCA1 protein sequences analyzed were obtained from NCBI.

Species	Protein	NCBI Accession	Sequence Length
Chimpanzee (*Pan troglodytes*)	Breast cancer type 1 susceptibility protein homolog	Q9GKK8.2	1863 aa
Gorilla (*Gorilla gorilla gorilla*)	Breast cancer type 1 susceptibility protein homolog	Q6J6I8.1	1863 aa
Sumatran orangutan (*Pongo abelii*)	Breast cancer type 1 susceptibility protein isoform X1	XP_003778880.3	1863 aa

All three selected proteins have a length of 1863 amino acids, indicating that the analyzed sequences correspond to proteins of the same overall size.

BLAST RESULTS

Species	Accession	Identity	Positives	Gaps	Bit Score	E-value
Chimpanzee	Q9GKK8.2	1834/1863 (98%)	1842/1863 (98%)	0/1863 (0%)	3773 bits	0.0
Gorilla	Q6J6I8.1	1826/1863 (98%)	1840/1863 (98%)	0/1863 (0%)	3757 bits	0.0
Pongo abelii	XP_003778880.3	1804/1863 (97%)	1823/1863 (97%)	0/1863 (0%)	3700 bits	0.0

- **Chimpanzee**

The chimpanzee BRCA1 sequence (Q9GKK8.2) showed 98% identity, with 1834 identical residues out of 1863. The alignment had 1842/1863 (98%) positives and no gaps. The alignment score was 3773 bits, with an E-value of 0.0.

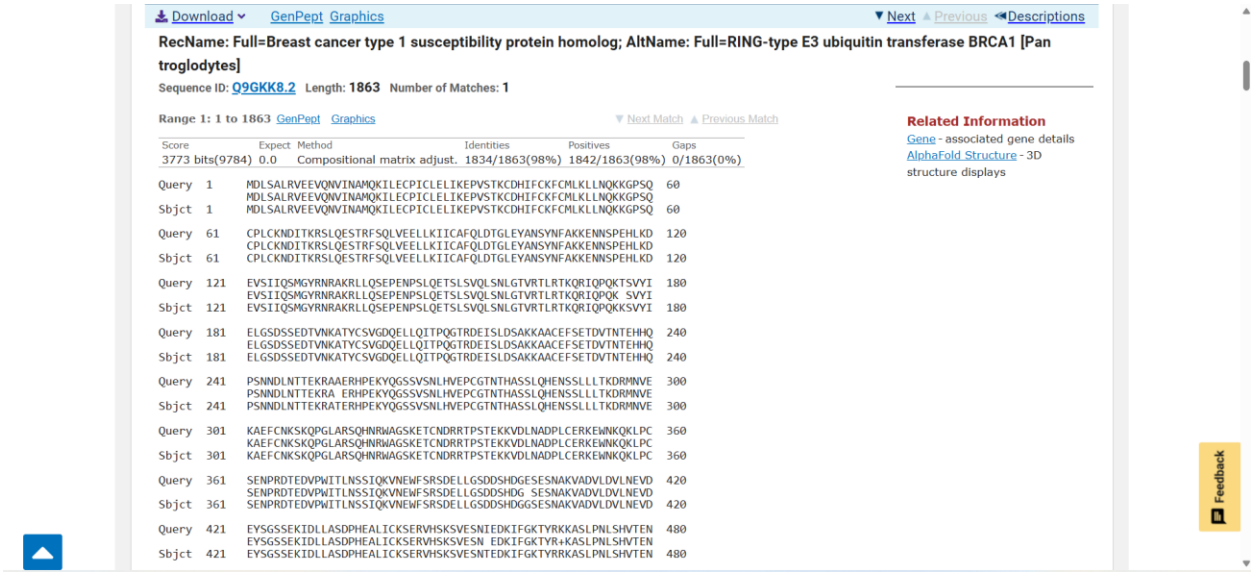
- **Gorilla**

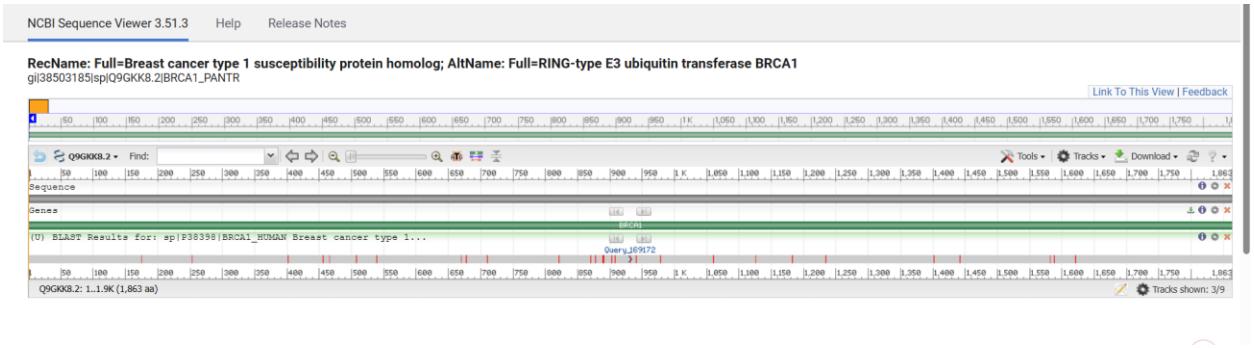
The gorilla BRCA1 sequence (Q6J6I8.1) showed 98% identity, with 1826 identical residues out of 1863. The alignment contained 1840/1863 (98%) positive residues and 0 gaps. The bit score was 3757 bits, and the E-value was 0.0.

- **Sumatran Orangutan**

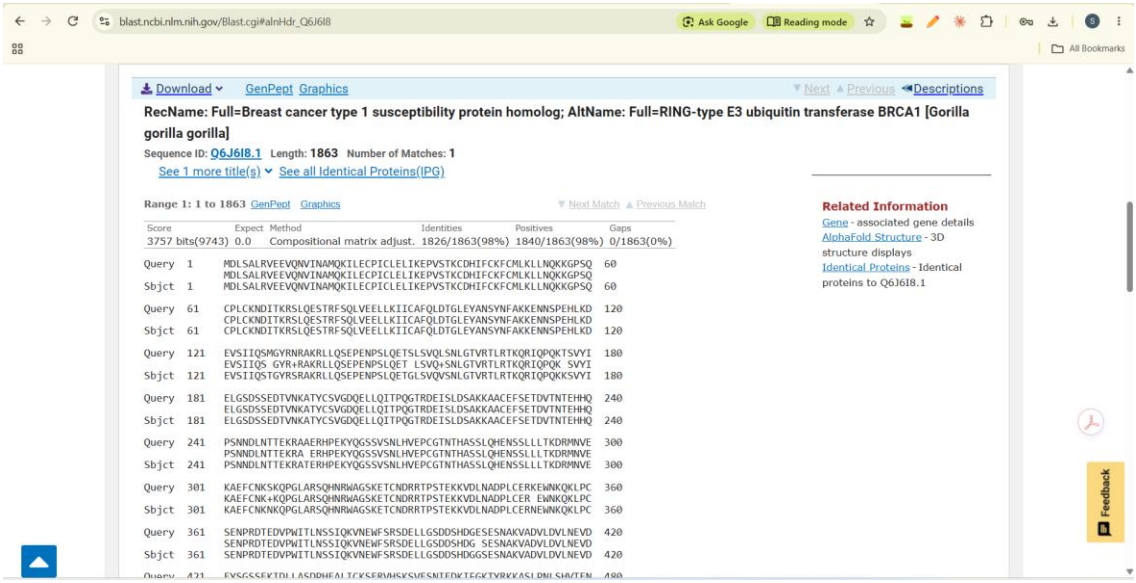
The Sumatran orangutan BRCA1 sequence (XP_003778880.3) showed 97% identity, with 1804 identical residues out of 1863. There were 1823/1863 (97%) positive residues and no gaps. The alignment score was 3700 bits, with an E-value of 0.0.

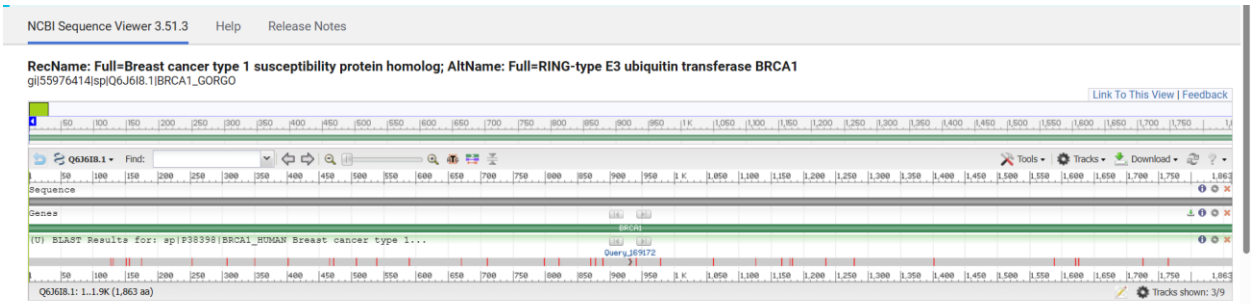
1. Accession no – Q9GKK8.2 results



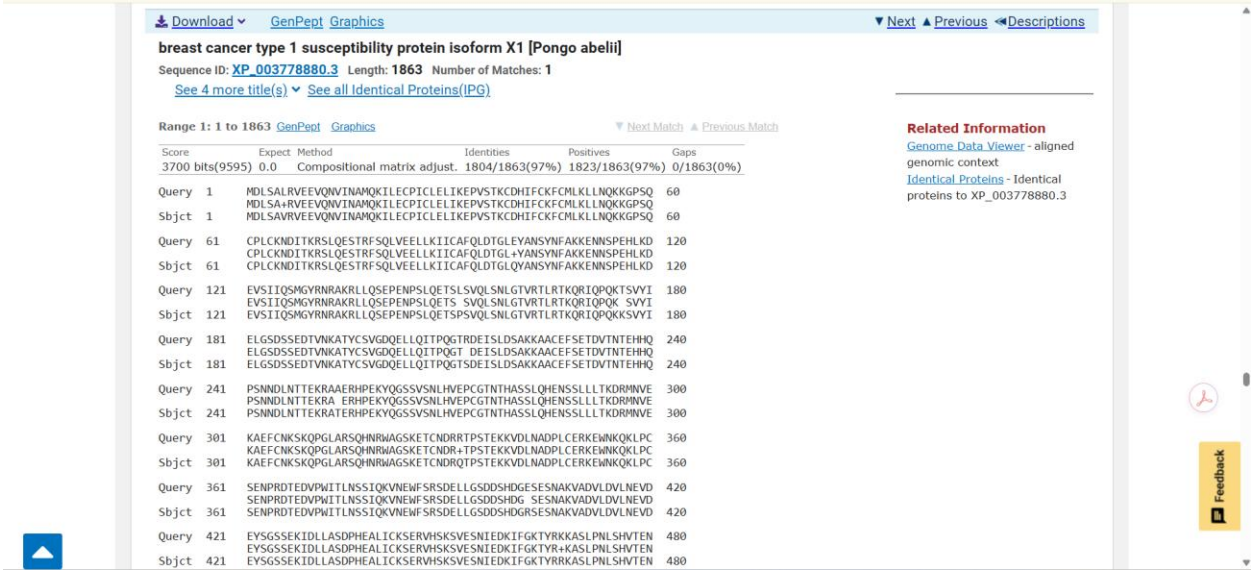


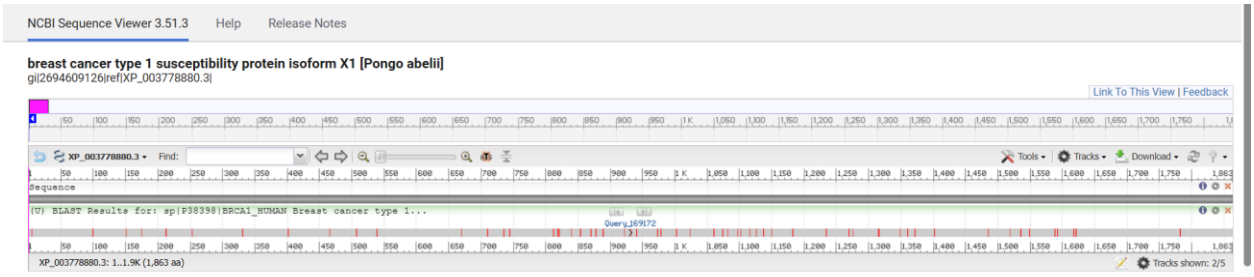
2. Accession no. – Q6J618.1 results





3. **Accession no. –** XP_003778880.3 results





ANALYSIS AND DISCUSSION

The BLAST results demonstrate that BRCA1 is highly conserved among chimpanzee, gorilla, and Sumatran orangutan.

The chimpanzee and gorilla sequences both showed 98% sequence identity with the reference sequence used in the BLAST analysis. The orangutan sequence showed slightly lower identity at 97%. Despite this small difference, all three sequences showed very high similarity.

The high number of identical amino acids indicates that most positions in the BRCA1 protein are conserved among these species. The positive values are also very high, ranging from 97% to 98%, indicating that even some amino acid substitutions are chemically similar and may therefore preserve protein structure or function.

Another important observation is that all three alignments contain 0 gaps across the complete 1863-amino-acid alignment. This indicates that no insertions or deletions were detected in the reported pairwise alignments. The complete alignment coverage further demonstrates that the sequences are highly similar throughout their full length rather than only within a small region.

The E-value for all three matches is 0.0, indicating that the observed similarities are extremely statistically significant and are not expected to occur by chance.

Overall, the results indicate strong evolutionary conservation of the BRCA1 protein among these closely related primates. The slightly greater sequence similarity observed for chimpanzee and gorilla compared with orangutan is consistent with the general pattern of closer evolutionary relationships among these species.

CONCLUSION

The BLAST analysis of BRCA1 protein sequences from chimpanzee, gorilla, and Sumatran orangutan demonstrated a very high degree of sequence conservation.

All three proteins are 1863 amino acids long and showed 97–98% sequence identity, 97–98% positives, and 0% gaps across the complete alignment. All BLAST comparisons had an E-value of 0.0, confirming that the similarities are highly significant.

The chimpanzee sequence showed the highest reported alignment score of 3773 bits, followed by the gorilla sequence at 3757 bits and the Sumatran orangutan sequence at 3700 bits.

These results demonstrate that the BRCA1 protein is strongly conserved among the three selected primate species. Such conservation is consistent with the important biological role of BRCA1 in cellular processes such as DNA damage response and maintenance of genomic stability.

REFERENCES

1. National Center for Biotechnology Information (NCBI). Protein database and BLAST resources.
2. NCBI Protein accession Q9GKK8.2 - BRCA1 homolog, *Pan troglodytes*.
3. NCBI Protein accession Q6J6I8.1 - BRCA1 homolog, *Gorilla gorilla gorilla*.
4. NCBI Protein accession XP_003778880.3 - BRCA1 isoform X1, *Pongo abelii*.
5. NCBI BLAST - Basic Local Alignment Search Tool for sequence similarity analysis.